Exercise 1

Consider split conformal prediction with data set $(X_1, Y_1), \cdots, (X_{n+1}, Y_{n+1}) \in \mathcal{X} \times \mathcal{Y}$ and the conformal risk control with given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. We write $S_i = s(X_i, Y_i)$ for all i for simplicity. Instead of what is in Chapter 10 of the book, we define, for all $\lambda \geqslant 0$:

$$\mathcal{C}_\lambda(x) = \{y \in \mathcal{Y} \mid s(x, y) < \lambda\}$$

and

$$L(y, \mathcal{C}_\lambda(x)) = \chi_{\{y \notin \mathcal{C}_\lambda(x)\}}.$$

Show that, with this definition, $L(y, \mathcal{C}_\lambda(x))$ is no longer always right continuous for every fixed $(x, y) \in \mathcal{X} \times \mathcal{Y}$. Next, given $\alpha \in (0, 1)$, give a concrete expression for

$$\hat{\lambda} = \inf \left\{ \lambda \geqslant 0 \mid \hat{R}(\lambda) \leqslant \alpha - \frac{\alpha - 1}{n} \right\}$$

in terms of the data, where

$$\hat{R}(\lambda) = \frac{1}{n} \sum_{i=1}^{n} L(Y_i, \mathcal{C}_\lambda(X_i)).$$

Show that, if we assume that $(S_1, \ldots, S_{n+1})$ is exchangeable and that the S_i's almost surely has no tie, then

$$P\left\{Y_{n+1} \in \mathcal{C}_{\hat{\lambda}}(X_{n+1})\right\} \geqslant 1 - \alpha.$$

Provide a counterexample to show that this need not hold if there are ties in S_i's with positive probability, even if they are exchangeable.

Solution Let $(x, y) \in \mathcal{X} \times \mathcal{Y}$ be given. If $\{\lambda_m\}_{m=1}^{\infty}$ is any decreasing sequence such that $\lambda_m > s(x, y)$ for every m and $\lambda_m \to s(x, y)$ as $m \to \infty$, then $L(y, \mathcal{C}_{\lambda_m}(x)) = 1$ for all m but $L(y, \mathcal{C}_{s(x,y)}(x)) = 0$, proving that $L(y, \mathcal{C}_\lambda(x))$ is not always right continuous.

We move to compute $\hat{\lambda}$. The definition of $\hat{R}(\lambda)$ and simple arithmetic rearrangement shows that, for any $\lambda \geqslant 0$,

$$\hat{R}(\lambda) \leqslant \alpha - \frac{\alpha - 1}{n}$$

if and only if

$$\sum_{i=1}^{n} \chi_{\{S_i < \lambda\}} \geqslant (n + 1)(1 - \alpha).$$

Thus, it is obvious that

$$\hat{\lambda} = S_{(\lceil (n+1)(1-\alpha) \rceil)}.$$

Here we have used order statistics $S_{(1)} \leqslant \cdots \leqslant S_{(n)}$ for the first n values of $S_1, \ldots, S_n$. Consequently:

$$C_{\hat{\lambda}}(X_{n+1}) = \left\{y \in \mathcal{Y} \mid s(X_{n+1}, y) < S_{(\lceil (n+1)(1-\alpha) \rceil)}\right\}.$$

To prove coverage, let $\alpha \in (0, 1)$ be given. Assume first that that $(S_1, \ldots, S_{n+1})$ is exchangeable and there is almost surely no tie in the S_i's. Then

$$P\left\{Y_{n+1} \in C_{\hat{\lambda}}(X_{n+1})\right\} = P\left\{S_{n+1} < S_{(\lceil (n+1)(1-\alpha) \rceil)}\right\} = \frac{\lceil (n+1)(1-\alpha) \rceil}{n+1} \geqslant 1-\alpha,$$

where the second equality uses both assumptions: when $S_{(1)} < \cdots < S_{(n)}$, inserting a new exchangeable point S_{n+1} to get exactly right before $\lceil (n+1)(1-\alpha) \rceil$th place has the probability $\frac{\lceil (n+1)(1-\alpha) \rceil}{n+1}$. If we remove the assumption that there are no ties, then everything can go wrong. For example, just setting $s \equiv 0$. Then $\hat{\lambda} = 0$ and $C_{\hat{\lambda}}(x) = \emptyset$ for all x, clearly not yielding any good coverage probability. ////